Generating self-signed certificates

Snapdrive for Unix

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The SnapDrive for UNIX daemon service requires that you generate a self-signed certificate for authentication. This authentication is required while communicating with the CLI.

Steps

1. Generate an RSA key:

   `$ openssl genrsa 1024 > host.key $ chmod 400 host.key`

   ```
   # openssl genrsa 1024 > host.key Generating
   RSA private key, 1024 bit long modulus
   ..............................++++++ ...++++++ e is 65537(0x10001)
   # chmod 400 host.key
   ```

2. Create the certificate:

   `$ openssl req -new -x509 -nodes -sha1 -days 365 -key host.key > host.cert`

   The `-new`, `-x509`, and `-nodes` options are used to create an unencrypted certificate. The `-days` option specifies the number of days the certificate remains valid.

3. When asked to fill out the certificate's x509 data, enter your local data:

   ```
   # openssl req -new -x509 -nodes -sha1 -days 365 -key host.key >
   host.cert
   You are about to be asked to enter information that will be incorporated
   into your certificate request.
   What you are about to enter is what is called a Distinguished Name or a
   DN. There are quite a few fields
   but you can leave some blank For some fields there will be a default
   value, If you enter '.', the field will be left blank.

   Country Name (2 letter code) [AU]:US
   State or Province Name (full name) [Some-State]:California
   Locality Name (eg, city) []:Sunnyvale
   Organization Name (eg, company) [Internet Widgits Pty Ltd]:abc.com
   Organizational Unit Name (eg, section) []:
   Common Name (eg, YOUR name) []:localhost
   Email Address []:postmaster@example.org
   ```

   The `Common Name` value must be *localhost*.

4. Extract metadata (optional).

```
$ openssl x509 -noout -fingerprint -text < host.cert > host.info
```

You can save the certificate metadata for quick reference later.

5. Combine key and certificate data.

   SnapDrive for UNIX requires the key and certificate data to be in the same file. The combined file must be protected as a key file.

   **$ cat host.cert host.key > host.pem **

   && rm host.key

   $ chmod 400 host.pem

```
# cat host.cert host.key >  /opt/NetApp/snapdrive.pem
# rm host.key rm: remove regular file `host.key'? y
# chmod 400 /opt/NetApp/snapdrive.pem
```

6. Add the complete path of the daemon certificate to the *`sdu-daemon-certificate-path`* variable of the `snapdrive.conf` file.

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